

CS4092 - Database Design and Development

Final Project Requirements Document: Online Restaurant & Delivery System

Course: Database Design and Development (CS4092) – **Author:** Sharanya Awasthi
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Backend: MySQL 8.0+ Relational Database

Interface: Python 3.12 Interactive CLI

1. Project Overview & Motivation

For this project, the goal is to build a practical application that models a real-world system used daily: an online restaurant ordering and delivery management system. In a fast-paced food service environment, managing customer profiles, delivery addresses, menu items, order tracking, and staff management can quickly become chaotic without a well-structured relational database backend. Secondly, I had worked on a similar python project in high school so I was familiar with how I could build my loops and if-statements in python for this to work.

The core objective is to design a MySQL database backend coupled with a Python command-line interface (CLI). The system cleanly separates two primary user roles:

- **Customers:** Register accounts, sign in, browse menu items, construct multi-item orders, and review detailed order history.
- **Staff Members:** Manage menu offerings, update live delivery statuses, and monitor operational sales statistics.

By structuring the architecture across six normalized tables with strict primary and foreign key constraints, the database maintains strict data integrity, eliminates duplicate registrations, calculates billing accurately, and delivers actionable business insights.

2. User Roles & Detailed Use Cases

A. Customer Use Cases

1. Customer Registration (Sign Up)

- **Description:** A new customer creates an account by submitting their full name, unique email address, and phone number.
- **Data Flow:** The application inserts the record into the Customers table, generates an auto-incremented customer_id, and immediately prompts for delivery address details (street address, city, state, zip code) stored in the Addresses table.
- **Outcome:** The customer is assigned a generated customer_id, which serves as their primary login key.

2. Customer Sign In

- **Description:** Existing customers authenticate using their unique customer_id.
- **Outcome:** The database validates the ID. Upon successful verification, the user enters the personalized Customer Portal.

3. Menu Browsing

- **Description:** Customers explore all currently available food and beverage options.
- **Outcome:** Displays a formatted menu listing item_id, item_name, category, and price.

4. Multi-Item Order Placement

- **Description:** Customers construct complete orders containing multiple distinct items within a single session.
- **Data Flow:** The user selects a saved address ID. An order record is created in Orders with initial status 'Placed'. In an interactive loop, items and quantities are recorded in OrderItems while running total costs update continuously until checkout.
- **Outcome:** The order commits to the database, returning the assigned order_id and final balance.

5. Order History & Tracking

- **Description:** Customers track real-time order status and review historical account purchases.
- **Outcome:** Outputs order IDs, timestamps, item breakdowns, total costs, and live statuses (Placed, Preparing, Out for Delivery, Completed).

B. Staff & Admin Use Cases

1. Staff Sign In & Registration

- **Description:** Staff authenticate via their assigned staff_id. Unregistered staff are prompted to enter their name and role for instant registration into the Staff table.

2. Menu Management (Add Items)

- **Description:** Authorized staff expand restaurant offerings by adding new menu items (Name, Category, Price, Availability).

3. Live Order Status Updates

- **Description:** Staff update order pipeline statuses in real-time ('Placed' → 'Preparing' → 'Out for Delivery' → 'Completed').

4. Sales Statistics & Business Analytics

- **Description:** Staff access interactive reporting queries to analyze performance:
- **Query 1:** Total gross revenue collected across all completed orders.
- **Query 2:** Identification of the most frequently purchased menu item.
- **Query 3:** High-value customer identification (customers with aggregate spend > \$50).
- **Query 4:** Targeted customer reporting (listing all customers who ordered 'Lobster Risotto').

3. Data Requirements & Database Schema Design

The database architecture consists of 6 core normalized relational tables:

Table Name	Primary Key / Unique Constraints	Attributes & Foreign Keys
Customers	customer_id (PK) email (UNIQUE)	full_name, email, phone_nb
Addresses	address_id (PK)	customer_id (FK -> Customers) address_label, street_address, city, state, zip_code
Staff	staff_id (PK)	full_name, staff_role
MenuItems	item_id (PK)	item_name, category, price, is_available
Orders	order_id (PK)	customer_id (FK -> Customers) address_id (FK -> Addresses) staff_id (FK -> Staff) order_date, order_status
OrderItems	order_item_id (PK)	order_id (FK -> Orders) item_id (FK -> MenuItems) quantity

4. Technical Requirements & Implementation Details

- **Database Engine:** MySQL 8.0+ Relational Database Management System (RDBMS).
- **Application Language:** Python 3.12 utilizing mysql.connector driver for database driver communication.
- **User Interface:** Interactive Command Line Interface (CLI) with clean sub-menus for role separation.
- **Data Integrity:** Enforced via Foreign Key relationships with ON DELETE CASCADE, UNIQUE constraints on customer emails, and explicit transactional boundaries (db.commit()).